Homework 08

Question 1: Boundary and margin

Answer (a)

Since $\|\beta\|_2 = 1$, the distance to the decision boundary is $|f(x_i)|$, and the functional margin is $y_i f(x_i)$.

```
d <- read.csv("data/margin-geometry-20.csv")

f <- 0.8*d$x1 + 0.6*d$x2
m <- d$y * f

result <- data.frame(
  Observation = d$observation,
  Distance = abs(f),
  Functional_Margin = m,
  Position = ifelse(abs(m - 1) < 1e-8, "On",
                    ifelse(m < 1, "Inside", "Beyond"))
)

knitr::kable(result, digits = 4)
```

Observation	Distance	Functional_Margin	Position
1	1.00	1.00	On
2	1.35	1.35	Beyond
3	1.50	1.50	Beyond
4	1.70	1.70	Beyond
5	1.90	1.90	Beyond
6	2.20	2.20	Beyond
7	2.40	2.40	Beyond

Observation	Distance	Functional_Margin	Position
8	0.64	0.64	Inside
9	0.20	0.20	Inside
10	0.20	-0.20	Inside
11	1.00	1.00	On
12	1.30	1.30	Beyond
13	1.55	1.55	Beyond
14	1.75	1.75	Beyond
15	1.95	1.95	Beyond
16	2.15	2.15	Beyond
17	2.45	2.45	Beyond
18	0.64	0.64	Inside
19	0.24	0.24	Inside
20	0.20	-0.20	Inside

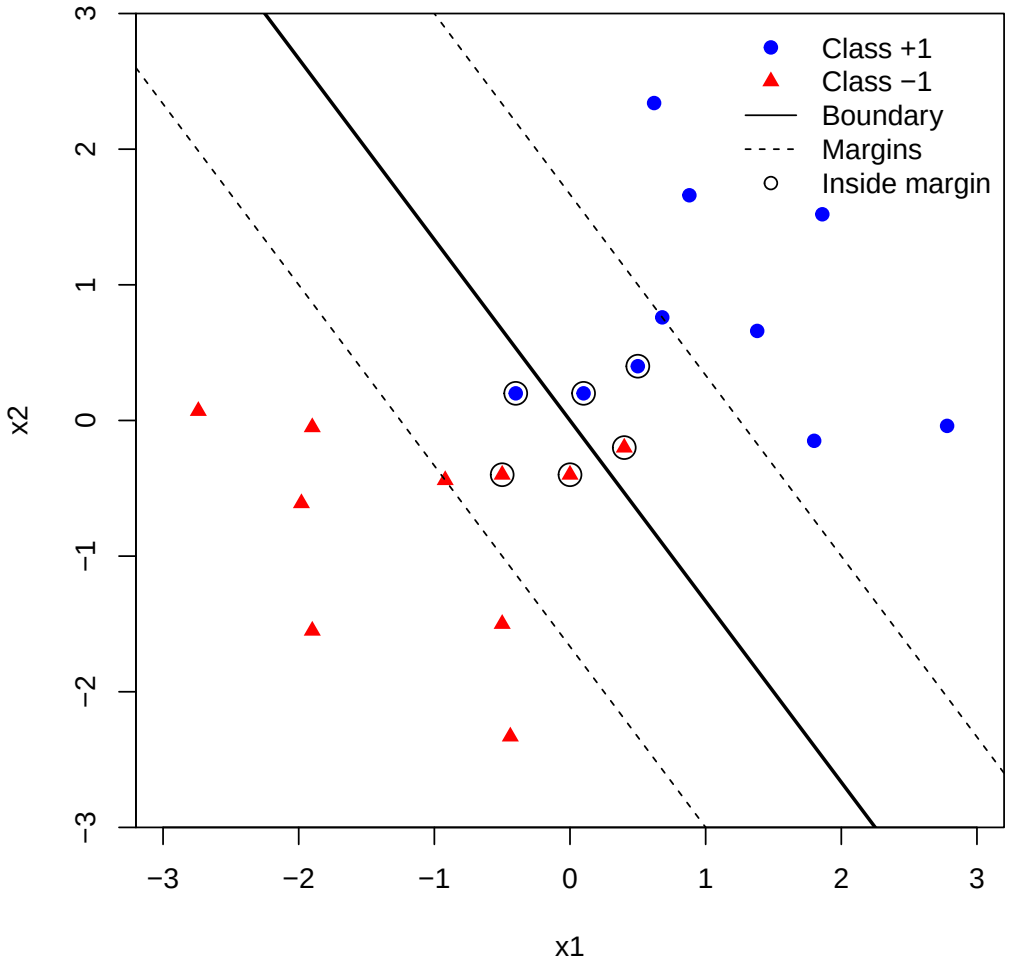
Answer (b)

```
par(pin = c(5.12, 4.8))
plot(d$x1, d$x2,
     col = ifelse(d$y == 1, "blue", "red"),
     pch = ifelse(d$y == 1, 19, 17),
     xlim = c(-3.2, 3.2), ylim = c(-3, 3),
     asp = 1, xaxs = "i", yaxs = "i", xlab = "x1", ylab = "x2")

abline(a = 0, b = -4/3, lwd = 2)
abline(a = 5/3, b = -4/3, lty = 2)
abline(a = -5/3, b = -4/3, lty = 2)

inside <- result$Position == "Inside"
points(d$x1[inside], d$x2[inside], pch = 1, cex = 1.8)

legend("topright",
       legend = c("Class +1", "Class -1", "Boundary",
                  "Margins", "Inside margin"),
       col = c("blue", "red", "black", "black", "black"),
       pch = c(19, 17, NA, NA, 1),
       lty = c(NA, NA, 1, 2, NA), bty = "n")
```



Answer (c)

```
table(result$Position)
```

```
Beyond Inside     On
    12      6      2
```

```r
d$observation[m < 0]
```

```
[1] 10 20
```

There are 6 observations inside the margin, 2 on the margin, and 12 beyond the margin. Observations 10 and 20 are misclassified because their functional margins $y_i f(x_i)$ are negative (both equal to -0.2).

Question 2: Margin in four dimensions

Answer (a): Geometric margin

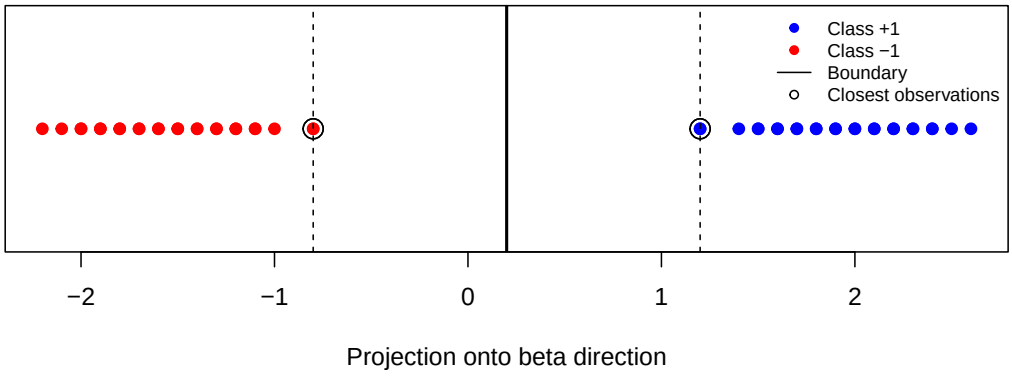
The coefficient vector is $\beta = (1, 2, -2, 4)^\mathsf{T}$, with $\|\beta\|_2 = 5$. Project each observation onto its unit direction:

$$t_i = \frac{\mathbf{x}_i^\mathsf{T} \beta}{5}.$$

The decision boundary is at $t = -\beta_0/5 = 0.2$.

```r
d4 <- read.csv("data/margin-geometry-4d.csv")
beta <- c(1, 2, -2, 4)
X4 <- as.matrix(d4[, c("x1", "x2", "x3", "x4")])
projection <- as.vector(X4 %*% beta) / sqrt(sum(beta^2))
boundary <- 1/5
M <- min(abs(projection - boundary))
closest <- abs(abs(projection - boundary) - M) < 1e-8

plot(projection, rep(0, length(projection)),
     col = ifelse(d4$y == 1, "blue", "red"),
     pch = 19, yaxt = "n", ylab = "", ylim = c(-0.5, 0.5),
     xlab = "Projection onto beta direction")
abline(v = boundary, lwd = 2)
abline(v = boundary + c(-M, M), lty = 2)
points(projection[closest], rep(0, sum(closest)), pch = 1, cex = 1.8)
legend("topright",
       legend = c("Class +1", "Class -1", "Boundary", "Closest observations"),
       col = c("blue", "red", "black", "black"),
       pch = c(19, 19, NA, 1), lty = c(NA, NA, 1, NA),
       bty = "n", cex = 0.75)
```




```
M
```

```
[1] 1
```

The closest projected observations are at $t = -0.8$ and $t = 1.2$, each a distance 1 from the boundary at $t = 0.2$. Thus,

$$\boxed{M = \min_i |t_i - 0.2| = 1}.$$

Answer (b): SVM normalization

```
f4 <- -1 + as.vector(X4 %*% beta)
min_margin <- min(d4$y * f4)
min_margin
```

```
[1] 5
```

```
beta_star <- beta / min_margin
beta0_star <- -1 / min_margin
beta_star
```

```
[1]  0.2  0.4 -0.4  0.8
```

```
beta0_star
```

```
[1] -0.2
```

```
min(d4$y * (beta0_star + as.vector(X4 %*% beta_star)))
```

```
[1] 1
```

The minimum functional margin is $\min_i y_i f(\mathbf{x}_i) = 5$. Dividing all coefficients by 5 preserves the decision boundary and sets the smallest functional margin to 1:

$$\boxed{\beta^* = (0.2, 0.4, -0.4, 0.8)^\top}, \qquad \boxed{\beta_0^* = -0.2}.$$

The normalized decision boundary is

$$-0.2 + 0.2x_1 + 0.4x_2 - 0.4x_3 + 0.8x_4 = 0.$$

Question 3: Peer review of a Homework 7 skill

Answer

GitHub Issue: Peer review

Overall Assessment

The overall performance is good. In the original report, the PCA-LDA model was selected, yielding a test error rate of 0.252. In my test run, standard LDA was chosen, resulting in a test error rate of 0.324. This Skill successfully guides the AI through LDA/QDA comparison, cross-validation, model selection, and test set evaluation.

Improvement 1: Clarify rules for resolving ties in model selection

In instruction 5 of the submitted Skill, the author requires "breaking ties in favor of the simpler model." However, the Skill does not define how complexity should be compared among ordinary LDA, QDA, PCA-based models, and probability blends. It also does not specify whether ties mean exactly equal cross-validation errors or errors within a numerical tolerance.

This could lead to inconsistent selections when cross-validation errors are tied. No tie occurred in my run, so this is a potential ambiguity rather than an observed selection error.

I recommend explicitly defining how to compare the complexity of tied candidates, including PCA-based and blended models. The Skill should also clarify whether ties require exactly equal errors or allow a specified numerical tolerance.

Improvement 2: Clarify how to handle model failures during cross-validation

The Skill requirements mandate logging failures, yet there are no specific rules regarding how to handle cases where a model fails on only a subset of folds. This creates potential ambiguity in execution.

As an improvement, I believe that rules for re-running the model after a failure should be established in advance. If the model still fails to complete all folds, it should be excluded from the pool of comparable candidate models, and the reason for the exclusion should be logged.

Improvement 3: Preserve execution details for reproducibility

The third point concerns reproducibility and the documentation of the execution process as well. Currently, the Skill provides code only upon request. To facilitate the reproduction of the execution process by others, the Skill could be configured to save—by default—the code, data splits, cross-validation fold indices, candidate parameter grids, and software versions. This does not imply saving vast amounts of unnecessary code by default; rather, it means preserving sufficient information to enable others to accurately reproduce the model selection and evaluation processes.